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The least common multiple of polynomial values over function fields

by

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Abstract. Cilleruelo conjectured that for an irreducible polynomial $f \in \mathbb{Z}[X]$ of degree $d \geq 2$ one has

$$\log \operatorname{lcm}(f(1), \dots, f(N)) \sim (d-1)N \log N$$

as $N \to \infty$. He proved it in the case $d = 2$ but it remains open for every polynomial with $d > 2$.

We investigate the function field analogue of the problem by considering polynomials over the ring $\mathbb{F}_q[T]$. We state an analogue of Cilleruelo's conjecture in this setting: denoting

$$L_f(n) := \operatorname{lcm}(f(Q) : Q \in \mathbb{F}_q[T] \text{ monic}, \deg Q = n)$$

we conjecture that

$$\deg L_f(n) \sim c_f(d-1)nq^n, \quad n \to \infty \tag{1}$$

(c_f is an explicit constant depending only on f, typically $c_f = 1$). We give both upper and lower bounds for $L_f(n)$ and show that the asymptotic (1) holds for a class of "special" polynomials, initially considered by Leumi in this context, which includes all quadratic polynomials and many other examples as well. We fully classify these special polynomials. We also show that $\deg L_f(n) \sim \deg \operatorname{rad} L_f(n)$ (in other words, the corresponding LCM is close to being squarefree), which is not known over $\mathbb{Z}$.

1. Introduction. While studying the distribution of prime numbers, Chebyshev estimated the least common multiple of the first N integers. This was an important step towards the Prime Number Theorem. In fact the condition $\log \operatorname{lcm}(1, \dots, N) \sim N$ is equivalent to the Prime Number Theorem. This problem later inspired a more general problem of studying the least common multiple of polynomial sequences. For a linear polynomial $f(X) = kX + h$, $k, h \in \mathbb{Z}$, $k > 0, h + k > 0$ it was observed by Bateman

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and Kalb [BKS02] that $\log \operatorname{lcm}(f(1), \dots, f(N)) \sim c_f N$ as $N \to \infty$, where $c_f = \frac{k}{\varphi(k)} \sum_{1 \le m \le k,\, (m,k)=1} \frac{1}{m}$, which is a consequence of the Prime Number Theorem for arithmetic progressions.

Cilleruelo [Cil21] conjectured that for any irreducible polynomial $f \in \mathbb{Z}[X]$ of degree $d \ge 2$ the following estimate holds:

$$\log \operatorname{lcm}(f(1), \dots, f(N)) \sim (d-1) N \log N \tag{1.1}$$

as f is fixed and $N \to \infty$.

Convention. Throughout the rest of the paper, in all asymptotic notation the polynomial f, and the parameter q to appear later, are assumed fixed, while the parameter N (or n) is taken to infinity. If any parameters other than f, q, N, n appear in the notation, the implied constant or rate of convergence are uniform in these parameters.

Cilleruelo proved (1.1) for quadratic polynomials and there are no known examples of polynomials of degree $d > 2$ for which the conjecture is known to hold. Cilleruelo's argument also shows the predicted upper bound, i.e. if $f \in \mathbb{Z}[X]$ is an irreducible polynomial of degree $d \ge 2$, then

$$\log \operatorname{lcm}(f(1), \dots, f(N)) \lesssim (d-1) N \log N.$$

Maynard and Rudnick [MR21] provided a lower bound of the correct order of magnitude:

$$\log \operatorname{lcm}(f(1), \dots, f(N)) \gtrsim \frac{1}{d} N \log N. \tag{1.2}$$

Sah [Sah20] improved upon this lower bound while also providing a lower bound for the radical of the least common multiple:

$$\log \operatorname{lcm}(f(1), \dots, f(N)) \gtrsim N \log N,$$

and

$$\log \operatorname{rad} \operatorname{lcm}(f(1), \dots, f(N)) \gtrsim \frac{2}{d} N \log N. \tag{1.3}$$

Rudnick and Zehavi [RZ20] established an averaged form of (1.1) with f varying in a suitable range. Leumi [Leu21] studied a function field analogue of the problem. In the present work we expand upon and generalize the results and conjectures in [Leu21], as well as correct some erroneous statements and conjectures from the latter work. Despite some overlap, we have kept our exposition self-contained and independent of [Leu21].

1.1. The function field analogue. Let $q = p^k$ be a prime power. For a polynomial $f \in \mathbb{F}_q[T][X]$ of degree $d \ge 1$ of the form

$$f(X) = f_d X^d + f_{d-1} X^{d-1} + \dots + f_0, \qquad f_i \in \mathbb{F}_q[T],$$

set

$$L_f(n) := \operatorname{lcm}(f(Q) : Q \in M_n), \tag{1.4}$$

where

$$M_n := \{Q \in \mathbb{F}_q[T] \text{ monic} : \deg Q = n\}.$$

Also denote

$$V_f := \{g \in \mathbb{F}_q[T] : f(X+g) = f(X)\}.$$

The set V_f is a finite-dimensional $\mathbb{F}_p$-linear subspace of $\mathbb{F}_q[T]$ (see Lemma 2.1 below). Denote

$$c_f := \frac{1}{|V_f|}.$$

We now state a function field analogue of Cilleruelo's conjecture.

CONJECTURE 1.1. *Let $f \in \mathbb{F}_q[T][X]$ be a fixed irreducible polynomial with $\deg_X f = d \geq 2$. Then*

$$\deg L_f(n) \sim c_f(d-1)nq^n, \qquad n \to \infty.$$

REMARK 1.2. The expression $(d-1)nq^n$ is directly analogous to the product $(d-1)N\log N$ appearing in (1.1). Over the integers (or generally in characteristic 0), if f is not constant then V_f is always trivial and no constant c_f appears in (1.1). This is because if $0 \neq g \in V_f$ then $2g, \ldots, dg$ are also in V_f, implying $f(0) = f(g) = \cdots = f(dg)$. Since $f(x) - f(0)$ has at most d roots, we reach a contradiction. Even over $\mathbb{F}_q[T]$, the typical case that occurs for "most" polynomials is $c_f = 1$ (i.e. V_f is trivial). A heuristic justification of Conjecture 1.1 will be given in Section 4.

In the present paper we prove that Conjecture 1.1 gives the correct upper bound:

THEOREM 1.3.

$$\deg L_f(n) \lesssim c_f(d-1)nq^n.$$

The proof of Theorem 1.3 will be given in Section 3. We also give a lower bound of the correct order of magnitude (this bound is comparable to the bound in [Sah20, Theorem 1.3] under a mild assumption on f):

THEOREM 1.4.

(i) *We have*

$$\deg L_f(n) \gtrsim \frac{d-1}{d} nq^n.$$

(ii) *If $p \nmid d$ or $f_d \nmid f_{d-1}$ then*

$$\deg L_f(n) \gtrsim nq^n.$$

The proof of Theorem 1.4 will be given in Section 5. We note that by Lemma 2.1 below we always have $c_f \geq 1/d$ and $c_f \geq 1/(d-1)$ if $p \nmid d$ or $f_d \nmid f_{d-1}$, so this is consistent with Conjecture 1.1.

Regarding the radical of the LCM,

$$\ell_f(n) := \operatorname{rad} \operatorname{lcm}(f(Q) : Q \in M_n),$$

we prove the following

Theorem 1.5.

$$\deg \ell_f(n) \sim \deg L_f(n).$$

The proof of Theorem 1.5 will be given in Section 6. As a consequence, our lower bounds for $L_f(n)$ apply also to $\ell_f(n)$. The analogous statement over $\mathbb{Z}$ is not known and the best lower bound over $\mathbb{Z}$, namely (1.3), has a smaller constant (if $d > 2$) than the best known lower bound for $L_f(N)$ given by (1.2). The key ingredient in the proof which is unavailable over $\mathbb{Z}$ is the work of Poonen [Poo02] on squarefree values of polynomials over $\mathbb{F}_q[T]$ (later generalized by Lando [Lan15] and Carmon [Car21]). If one assumes the ABC conjecture, then the analogous statement $\log \ell_f(N) \sim \log L_f(N)$ can be proven over $\mathbb{Z}$ (see Remark 6.2 below).

For a class of polynomials $f \in \mathbb{F}_q[T][X]$ that we call *special* (first introduced by Leumi [Leu21]) it is possible to establish Conjecture 1.1 in full. This class includes all quadratic polynomials, but also many polynomials of higher degree. We now define special polynomials over an arbitrary unique factorization domain (UFD). This definition was introduced in [Leu21].

Definition 1.6. A polynomial $f \in R[X]$ of degree $d = \deg f \geq 2$ is called *special* if the bivariate polynomial $f(X) - f(Y)$ factors into a product of linear terms in $R[X, Y]$:

$$f(X) - f(Y) = \prod_{i=1}^{d} (a_i X + b_i Y + c_i), \qquad a_i, b_i, c_i \in R.$$

Example 1.7.

(i) A quadratic polynomial is always special because

$$AX^2 + BX + C - (AY^2 + BY + C) = (X - Y)(AX + AY + B).$$

(ii) If $R = \mathbb{F}_p$ then $f = X^p$ is special because

$$X^p - Y^p = \prod_{a \in \mathbb{F}_p} (X - aY).$$

For a special polynomial f, Conjecture 1.1 can be established in full.

Theorem 1.8. *If $f \in \mathbb{F}_q[T][X]$ is irreducible and special then Conjecture 1.1 holds for f.*

The proof of Theorem 1.8 will be given in Section 7.

EXAMPLE 1.9. The polynomial $f = X^p - T \in \mathbb{F}_p[T][X]$ is irreducible (since it is linear in T) and special (similarly to Example 1.7(ii)). Hence Conjecture 1.1 holds for it.

We fully classify the set of special polynomials over an arbitrary UFD R.

THEOREM 1.10. *Let R be a UFD, K its field of fractions and $p = \mathrm{char}(K)$.*

(i) *Assume $p = 0$. Then $f \in R[X]$ is special iff it is of the form*

$$f(X) = f_d(X + A)^d + C, \quad 0 \neq f_d \in R,\ A, C \in K,$$

where $d \geq 2$ is such that there exists a primitive dth root of unity in K.

(ii) *Assume $p > 0$. Then $f \in R[X]$ of degree $\deg f = d = p^l m \geq 2$, $(m, p) = 1$ is special iff it is of the form*

$$f(X) = f_d \prod_{i=1}^{p^v} (X - b_i + A)^{mp^{l-v}} + C,$$
$$0 \neq f_d \in R, \quad A, C \in K, \quad 0 \leq v \leq l,$$
$$\zeta \in K, \quad V = \{b_1, \ldots, b_{p^v}\} \subset K,$$

where ζ is a primitive mth root of unity and V is an $\mathbb{F}_p(\zeta)$-linear subspace of K with $|V| = p^v$.

The proof of Theorem 1.10 will be given in Section 8.

We now briefly discuss how our main conjecture and results compare with the work of Leumi [Leu21], which also studied the function field analogue of Cilleruelo's conjecture and influenced the present work. First, [Leu21] states Conjecture 1.1 without the constant c_f. This is certainly false by Theorem 1.3 because it can happen that $c_f < 1$ (see Example 2.2 below). It seems to have been overlooked that if $g \in V_f$ and $Q \in M_n$, $n > \deg g$, then since $f(Q + g) = f(Q)$, the value $f(Q + g)$ contributes nothing new to the LCM on the RHS of (1.4) over the contribution of $f(Q)$. Once one accounts for this redundancy with the constant $c_f = 1/|V_f|$, Cilleruelo's original heuristic carries over to the function field case giving rise to Conjecture 1.1. Second, all results and conjectures in [Leu21] are stated only for an absolutely irreducible and separable $f \in \mathbb{F}_q[T][X]$. We do not make these restrictions here and it takes additional technical work to treat the general case. Third, the lower bound on $L_f(n)$ given in [Leu21] has a smaller constant than ours (thus the bound is weaker), comparable to the RHS of (1.3), and a lower bound for the radical of $L_f(n)$ is not stated explicitly. Fourth, in [Leu21] our Theorem 1.8 is stated without the constant c_f, which is incorrect in general for the same reasons explained above, although the arguments therein are essentially correct in the case $c_f = 1$. Finally, [Leu21] gives a classification of special

polynomials only in the case $p \nmid d$ (and it is stated only for the ring $\mathbb{F}_q[T]$), whereas we treat the general case.

2. Preliminaries. For background on the arithmetic of function fields see [Ros02]. For background on resultants, which we will use below, see [Lan02, §IV.8].

2.1. Notation. We now introduce some notation which will be used throughout Sections 2–7. Let p be a prime, $q = p^k$. For $Q \in \mathbb{F}_q[T]$ we denote by $|Q| = q^{\deg Q}$ the standard size of Q. For $P \in \mathbb{F}_q[T]$ prime and $Q \in \mathbb{F}_q[T]$ we denote by $v_P(Q)$ the exponent of P in the prime factorization of Q.

We will always fix a polynomial

$$f(X) = \sum_{i=0}^{d} f_i X^i \in \mathbb{F}_q[T][X], \qquad f_d \neq 0,$$

of degree d.

We also adopt the following conventions about notation:

- For a polynomial $Q \in \mathbb{F}_q[T]$ we denote by $\deg Q$ its degree in T. For a polynomial $g \in \mathbb{F}_q[T][X]$ we denote by $\deg g = \deg_X g$ its degree in X.
- For a polynomial $g \in \mathbb{F}_q[T][X]$ we denote by $g' = \frac{\partial g}{\partial X}$ its X-derivative. The T-derivative will be written explicitly $\frac{\partial g}{\partial T}$.
- $g \in \mathbb{F}_q[T][X]$ is called *separable* if it is separable as a polynomial in X, equivalently $g \notin \mathbb{F}_q[T][X^p]$.
- For $g, h \in \mathbb{F}_q[T][X]$ we denote by $\mathrm{Res}(g,h) = \mathrm{Res}_X(g,h)$ their resultant in X.

2.2. The space V_f. Recall that $V_f = \{g \in \mathbb{F}_q[T] : f(X+g) = f(X)\}$.

LEMMA 2.1. *Assume $d \geq 1$.*

(i) $|V_f| \leq d$.
(ii) V_f *an* $\mathbb{F}_p$*-linear subspace of* $\mathbb{F}_q[T]$.
(iii) $|V_f|$ *is a power of* p.
(iv) *If* $p \nmid d$ *then* V_f *is trivial.*
(v) *If* $f_d \nmid f_i$ *for some* $1 \leq i \leq d-1$ *then* $|V_f| \leq d-1$.

Proof. (i) Assume by way of contradiction that $|V_f| \geq d+1$. Then $f(g) = f(0)$ for every $g \in V_f$. Since $f(x) - f(0)$ has at most d roots, $f(x) = f(0)$ and f is constant, a contradiction.

(ii) It is obvious that $0 \in V_f$. Now let $a, b \in V_f$ and let us prove that $\alpha a + \beta b \in V_f$ where $\alpha, \beta \in \{0, 1, \ldots, p-1\}$. Recursively applying $f(X+a) =$

$f(X+b)=f(X)$ we obtain

$$f(X+\alpha a+\beta b)=f\Big(X+\sum_{i=1}^{\alpha}a+\sum_{i=1}^{\beta}b\Big)=f(X).$$

(iii) Obvious from (i) and (ii).

(iv) Let $g\in V_f$, so $f(X+g)=f(X)$. Comparing the coefficients of X^{d-1} we find $dg=0$. If $p\nmid d$ then $g=0$, so in this case $V_f=\{0\}$ and the claim follows.

(v) If $|V_f|=d$ then since all elements $g\in V_f$ are roots of $f(X)-f(0)$, we have $f=f_d\prod_{g\in V_f}(X-g)+f(0)$ and $f_d\,|\,f_i$ for all $1\le i\le d-1$, contradicting the assumption. ∎

Example 2.2 (Example of a polynomial with $|V_f|=d=\deg f$). Let $V=\{b_1,\dots,b_{p^v}\}\subset\mathbb{F}_q[T]$ be an $\mathbb{F}_p$-linear subspace and $C\in\mathbb{F}_q[T]$. Then the polynomial

$$f(X)=\prod_{i=1}^{p^v}(X-b_i)+C$$

has $V_f=V$. Thus $|V_f|=|V|=p^v=d$.

2.3. Roots of f modulo prime powers. In this subsection we study the quantity

$$\rho_f(P^k):=|\{Q \bmod P^k : f(Q)\equiv 0 \pmod P^k\}|,$$

i.e. the number of roots of f modulo a prime power P^k.

Lemma 2.3. *Let $g,h\in\mathbb{F}_q[T][X]$ be polynomials and let $P\in\mathbb{F}_q[T]$ be prime. If g,h have a common root modulo P^m then $P^m\,|\,R:=\mathrm{Res}(g,h)$.*

Proof. We can express R as $R=a(X)f(X)+b(X)g(X)$ for some $a(X)$, $b(X)\in\mathbb{F}_q[T][X]$ (see [Lan02, §IV.8]). Therefore, if there exists a $Q\in\mathbb{F}_q[T]$ such that $P^m\,|\,f(Q)$ and $P^m\,|\,g(Q)$, then P^m must also divide $a(Q)f(Q)+b(Q)g(Q)=R$. ∎

The proof of the next two lemmas is similar to the analogous proof for the integer case in [Nag21, proof of Theorem II].

Lemma 2.4. *Assume f is separable and denote $R:=\mathrm{Res}_X(f,f')\neq 0$. Denote $\mu=v_P(R)$. Let $x_0,x_1\in\mathbb{F}_q[T]$ be such that*

(a) $f(x_0)\equiv 0 \pmod{P^{\mu+1}}$;
(b) $f(x_1)\equiv 0 \pmod{P^{\beta+1}}$, *where* $\beta:=v_P(f'(x_0))$;
(c) $x_1\equiv x_0 \pmod{P^{\mu+1}}$.

Then $\beta\le\mu$ and $v_P(f'(x_1))=\beta$.

Proof. Since $f(x_0) \equiv 0 \pmod{P^{\mu+1}}$ and $P^{\mu+1} \nmid R$, by Lemma 2.3 we must have $P^{\mu+1} \nmid f'(x_0)$. Hence $\beta \leq \mu$. Now writing $x_1 = x_0 + tP^{\mu+1}$ with $t \in \mathbb{F}_q[T]$ and using $\beta \leq \mu$ and conditions (a) and (b) we have

$$f'(x_1) = f'(x_0 + tP^{\mu+1}) \equiv f'(x_0) \equiv 0 \pmod{P^{\beta}}$$

and

$$f'(x_1) = f'(x_0 + tP^{\mu+1}) \equiv f'(x_0) \not\equiv 0 \pmod{P^{\beta+1}},$$

so $v_P(f'(x_1)) = \beta$. ∎

LEMMA 2.5. *In the setup of Lemma* 2.4 *let* $\alpha > \mu$ *be an integer and assume that* $f(x_1) \equiv 0 \pmod{P^{\alpha+\beta}}$. *Set*

$$S_1 := \{x_1 + uP^{\alpha} : u \in \mathbb{F}_q[T]/P^{\beta}\} \subset \mathbb{F}_q[T]/P^{\alpha+\beta}.$$

Then

(i) *the elements of* S_1 *are roots of* f *modulo* $P^{\alpha+\beta}$;
(ii) *the number of roots of* f *modulo* $P^{\alpha+\beta+1}$ *that reduce modulo* $P^{\alpha+\beta}$ *to an element of* S_1 *is equal to* $|S_1| = q^{\beta \deg P}$.

Proof. To prove (i) we note that

$$f(x_1 + uP^{\alpha}) \equiv f(x_1) + uP^{\alpha} f'(x_1) \pmod{P^{2\alpha}}.$$

Thus $P^{\alpha+\beta} \mid f(x_1+uP^{\alpha})$ as $P^{\alpha+\beta} \mid f(x_1)$ and by Lemma 2.4 we have $P^{\beta} \mid f'(x_1)$ and $\alpha + \beta \leq \alpha + \mu < 2\alpha$.

To show (ii), consider the set of possible lifts from S_1 to $\mathbb{F}_q[T]/P^{\alpha+\beta+1}$, that is,

$$S_2 := \{x_1 + uP^{\alpha} + vP^{\alpha+\beta} : u \in \mathbb{F}_q[T]/P^{\beta},\ v \in \mathbb{F}_q[T]/P\}.$$

We will now determine for which u, v the element $x_1 + uP^{\alpha} + vP^{\alpha+\beta}$ is a root of f modulo $P^{\alpha+\beta+1}$. Using $2\alpha > \alpha + \beta$ we have

$$f(x_1 + uP^{\alpha} + vP^{\alpha+\beta}) \equiv f(x_1) + uP^{\alpha} f'(x_1) + vP^{\alpha+\beta} f'(x_1) \pmod{P^{\alpha+\beta+1}}.$$

Hence $x_1 + uP^{\alpha} + vP^{\alpha+\beta}$ is a root of f modulo $P^{\alpha+\beta+1}$ iff

$$f(x_1) + uP^{\alpha} f'(x_1) + vP^{\alpha+\beta} f'(x_1) \equiv 0 \pmod{P^{\alpha+\beta+1}}. \tag{2.1}$$

As $P^{\alpha+\beta} \mid f(x_1)$ and $v_P(f'(x_1)) = \beta$ (Lemma 2.4), we have (2.1) iff

$$u \equiv -\left(\frac{f'(x_1)}{P^{\beta}}\right)^{-1} \left(\frac{f(x_1)}{P^{\alpha+\beta}} + vf'(x_1)\right) \pmod{P}$$

Thus we have $|P|$ possible values of v and, for each of these, $|P^{\beta}|/|P|$ possible values of u. Overall we have $|P^{\beta}| = q^{\beta \deg P} = |S_1|$ possible values of (u, v) and the assertion follows. ∎

LEMMA 2.6. *Assume that* f *is separable and* $R = \operatorname{Res}(f, f') \neq 0$. *Let* P *be a prime in* $\mathbb{F}_q[T]$ *and* $\mu = v_P(R)$. *Then* $\rho_f(P^{2\mu+k}) = \rho_f(P^{2\mu+1})$ *for all* $k \geq 1$.

Proof. For each root x_1 of f modulo $P^{2\mu+k}$ we can apply Lemma 2.5 with $\alpha = 2\mu + k - \beta$, where $\beta = v_P(f'(x_1))$ (the condition $\alpha > \mu$ holds because $\beta \le \mu$ by Lemma 2.4 applied with $x_0 = x_1$), and deduce that the number of roots of f modulo $P^{2\mu+k+1}$ equals the number of roots of f modulo $P^{2\mu+k}$, i.e. $\rho_f(P^{2\mu+k+1}) = \rho_f(P^{2\mu+k})$. The assertion now follows by induction on k. ∎

LEMMA 2.7. *Assume $f \in \mathbb{F}_q[T][X^p]$ is inseparable and irreducible. Then $U := \mathrm{Res}\big(f, \frac{\partial f}{\partial T}\big) \neq 0$.*

Proof. Assume by way of contradiction that $U = 0$. Then $f, \frac{\partial f}{\partial T}$ have a common factor, and since f is irreducible we have $f \mid \frac{\partial f}{\partial T}$. Comparing degrees in T we must have $\frac{\partial f}{\partial T} = 0$. This means that $f \in \mathbb{F}_q[T^p, X^p]$ is a pth power, contradicting its irreducibility. ∎

LEMMA 2.8. *Assume that f is inseparable and irreducible, and $P^m \nmid U := \mathrm{Res}\big(f, \frac{\partial f}{\partial T}\big) \in \mathbb{F}_q[T]$ for some $P \in \mathbb{F}_q[T]$ prime and $m \ge 1$. Then $\rho_f(P^k) = 0$ for every $k \ge m+1$.*

Proof. Assuming the existence of $Q \in \mathbb{F}_q[T]$ such that $P^m \mid f(Q)$ (if f has no roots modulo P^m we are done), we will now prove that $P^{m+1} \nmid f(Q)$. Since $P^m \mid f(Q)$, we know that $P^m \nmid \frac{\partial f}{\partial T}(Q)$; otherwise, by Lemma 2.3 we would have $P^m \mid U$, contradicting our assumption.

Since f is inseparable and irreducible, we have $f' = 0$ and therefore

$$\frac{\partial f(Q)}{\partial T} = \frac{\partial f}{\partial T}(Q) + \frac{\partial f}{\partial X}(Q)\frac{dQ}{dT} = \frac{\partial f}{\partial T}(Q).$$

If $P^{m+1} \mid f(Q)$, write $f(Q) = P^{m+1}C$ and then

$$\frac{\partial f(T, Q(T))}{\partial T} = \frac{\partial (P^{m+1}C)}{\partial T} = P^{m+1}\frac{\partial C}{\partial T} + (m+1)P^m\frac{\partial P}{\partial T}C.$$

Thus $P^m \mid \frac{\partial f(T,Q(T))}{\partial T} = \frac{\partial f}{\partial T}(Q)$, contradicting the above observation that $P^m \nmid \frac{\partial f}{\partial T}(Q)$. ∎

PROPOSITION 2.9. *Assume that f is irreducible. Then $\rho_f(P^m) \ll 1$ for all P prime and $m \ge 1$.*

Proof. Since $\mathbb{F}_q[T]/P$ is a field, we have $\rho_f(P) \le d$. Thus it remains to handle the case $m \ge 2$.

If f is inseparable then using Lemmas 2.8 and 2.7 we see that there are only finitely many pairs P, m such that $m \ge 2$ and $\rho_f(P^m) \neq 0$ (as they must satisfy $P^m \mid \mathrm{Res}\big(f, \frac{\partial f}{\partial T}\big) \neq 0$).

Now if f is separable then by Lemma 2.6 there are only finitely many pairs (P, m) such that $m \ge 2$ and $\rho_f(P^m) \neq \rho_f(P^{m-1})$. Denote these pairs

by $(P_i, m_i)_{i=1}^s$. Then for all primes P and all $m \geq 1$,

$$\rho_f(P^m) \leq \max\{d, \rho_f(P_i^{m_i})_{i=1}^s\} \ll 1. \blacksquare$$

LEMMA 2.10. *Let $f \in \mathbb{F}_q[T][X^p]$ be inseparable and irreducible. Then there exist a separable and irreducible $h \in \mathbb{F}_q[T][X]$ such that $\rho_f(P) = \rho_h(P)$ for all primes $P \in \mathbb{F}_q[T]$.*

Proof. Write $f(X) = h(X^{p^m})$ for some $m \geq 1$ such that $h \in \mathbb{F}_q[T][X] \setminus \mathbb{F}_q[T][X^p]$ is separable. For any prime P, the m-fold Frobenius automorphism of $\mathbb{F}_q[T]/P$ given by $x \mapsto x^{p^m}$ establishes a one-to-one correspondence between the roots of f modulo P and the roots of h modulo P. Hence $\rho_f(P) = \rho_h(P)$. $\blacksquare$

3. The upper bound. Throughout this section the polynomial f is assumed *irreducible*. To obtain the upper and lower bounds on $L_f(n)$ (recall that this quantity was defined in (1.4)) we set

$$P_f(n) := \prod_{Q \in M_n} f(Q)$$

and write

$$\deg L_f(n) = c_f \deg P_f(n) - (c_f \deg P_f(n) - \deg L_f(n)), \tag{3.1}$$

where $c_f = 1/|V_f|$.

LEMMA 3.1. $\deg P_f(n) = dnq^n + O(q^n)$.

Proof. For sufficiently large n and $Q \in M_n$ we have $\deg f(Q) = dn + \deg f_d$, hence

$$\begin{aligned}\deg P_f(n) &= \deg \prod_{Q \in M_n} f(Q) = \sum_{Q \in M_n} \deg f(Q) = \sum_{Q \in M_n} (dn + \deg f_d) \\ &= dnq^n + O(q^n). \blacksquare\end{aligned}$$

CONVENTION. Throughout the rest of the paper, P will always denote a prime of $\mathbb{F}_q[T]$ and $\sum_P$ (resp. $\prod_P$) will denote a sum (resp. product) over all monic primes of $\mathbb{F}_q[T]$. A sum of the form $\sum_{a \leq \deg P \leq b}$ is over all monic primes in the corresponding degree range (and the same for products).

To estimate $c_f \deg P_f(n) - \deg L_f(n)$ we write the prime decompositions of $L_f(n)$ and $P_f(n)$ as

$$L_f(n) = \prod_P P^{\beta_P(n)}, \qquad P_f(n) = \prod_P P^{\alpha_P(n)},$$

where the products are over all (monic) primes in $\mathbb{F}_q[T]$.

CONVENTION. Throughout Sections 3–7 we will always assume that n is large enough so that $f(Q) \neq 0$ for all $Q \in M_n$. Thus $L_f(n), P_f(n) \neq 0$ and $\alpha_P(n), \beta_P(n)$ are well-defined nonnegative integers.

We have

$$\alpha_P(n) = \sum_{Q \in M_n} v_P(f(Q)), \qquad \beta_P(n) = \max\{v_P(f(Q)) : Q \in M_n\} \tag{3.2}$$

(recall that $v_P(Q)$ is the exponent of P in the prime factorization of Q). Combining (3.1) with Lemma 3.1 we have

$$\deg L_f(n) = c_f d n q^n - \sum_P (c_f \alpha_P(n) - \beta_P(n)) \deg P + O(q^n).$$

LEMMA 3.2. *For sufficiently large n,*

$$\beta_P(n) \le c_f \alpha_P(n).$$

Proof. Denote by $Q_{\max} \in M_n$ an element such that $\beta_P(n) = v_P(f(Q_{\max}))$. Then for each element $g \in V_f$ we have $f(Q_{\max} + g) = f(Q_{\max})$ and hence $\beta_P(n) = v_P(f(Q_{\max})) = v_P(f(Q_{\max} + g))$. For n sufficiently large we have $Q_{\max} + g \in M_n$, so

$$|V_f| \beta_P(n) = \sum_{g \in V_f} v_P(f(Q_{\max} + g)) \le \sum_{Q \in M_n} v_P(f(Q)) = \alpha_P(n).$$

The assertion follows since $c_f = 1/|V_f|$. ∎

The next lemma is the main tool for estimating $\alpha_P(n), \beta_P(n)$. For its proof we introduce the following notation, which will be used later as well:

$$s_f(P^k, n) = |\{Q \in M_n : f(Q) \equiv 0 \pmod{P^k}\}|, \tag{3.3}$$

where P is prime and $k \ge 1$.

LEMMA 3.3. *Let $P \in \mathbb{F}_q[T]$ be a prime.*

(i) $\beta_P(n) = O\big(\frac{n}{\deg P}\big)$.

(ii) *If f is separable and $P \nmid \operatorname{Res}(f, f')$, then*

$$\alpha_P(n) = q^n \frac{\rho_f(P)}{|P| - 1} + O\bigg(\frac{n}{\deg P}\bigg).$$

(iii) *If f is inseparable and $P \nmid \operatorname{Res}\big(f, \frac{\partial f}{\partial T}\big)$, then*

$$\alpha_P(n) = q^n \frac{\rho_f(P)}{|P|} + O\bigg(\frac{n}{\deg P}\bigg).$$

(iv) *For the finitely many "bad" primes where neither* (ii) *nor* (iii) *holds we have*

$$\alpha_P(n) = O(q^n).$$

Proof. To prove (i) we note that since there exists $Q \in M_n$ such that $P^{\beta_P(n)} \mid f(Q)$ and $f(Q) \ne 0$, we get

$$\beta_P(n) \deg P \le \deg f(Q).$$

For sufficiently large n we have $\deg f(Q) = dn + \deg f_d$, thus

$$\beta_P(n) \ll \frac{n}{\deg P},$$

establishing (i).

Using the notation (3.3) we have

$$\alpha_P(n) = \sum_{Q \in M_n} v_P(Q) = \sum_{Q \in M_n} \sum_{\substack{k \geq 1 \\ P^k | f(Q)}} 1 = \sum_{k \geq 1} \sum_{\substack{Q \in M_n \\ P^k | f(Q)}} 1 = \sum_{k \geq 1} s_f(P^k, n).$$

Note that

$$s_f(P^k, n) = q^n \frac{\rho_f(P^k)}{|P^k|} + O(1),$$

since if Q is a root of f modulo P^k and $\deg P^k \leq n$ then there are exactly $q^n/|P|^k$ elements in M_n that reduce to Q modulo P^k. And if $\deg P^k > n$ then $s_f(P^k, n) \leq \rho_f(P^k) \ll 1$ by Proposition 2.9. Hence

$$\begin{aligned} \alpha_P(n) &= \sum_{k \geq 1} s_f(P^k, n) = \sum_{k \geq 1} \left[q^n \frac{\rho_f(P^k)}{|P|^k} + O(1) \right] \\ &= q^n \sum_{k \geq 1} \frac{\rho_f(P^k)}{|P|^k} + O\left(\frac{n}{\deg P} \right). \end{aligned}$$

Let us look at the different cases:

- If f is separable and $P \nmid \operatorname{Res}(f, f')$ then we have $\rho_f(P^k) = \rho_f(P)$ by Lemma 2.6 and thus

$$\begin{aligned} \alpha_P(n) &= q^n \sum_{k \geq 1} \frac{\rho_f(P)}{|P|^k} + O\left(\frac{n}{\deg P} \right) \\ &= q^n \rho_f(P) \sum_{k \geq 1} \left(\frac{1}{|P|} \right)^k + O\left(\frac{n}{\deg P} \right) \\ &= q^n \frac{\rho_f(P)}{|P| - 1} + O\left(\frac{n}{\deg P} \right). \end{aligned}$$

- If f is inseparable and $P \nmid \operatorname{Res}_X\left(f, \frac{\partial f}{\partial T}\right)$ then using Lemma 2.8 we get

$$\alpha_P(n) = q^n \frac{\rho_f(P)}{|P|} + O\left(\frac{n}{\deg P} \right).$$

- For a general irreducible f, using the fact that $\rho_f(P^k) \ll 1$ for every $k \geq 1$

(Proposition 2.9), we get

$$\alpha_P(n) = q^n \sum_{k\geq 1} \frac{\rho_f(P^k)}{|P|^k} + O\Big(\frac{n}{\deg P}\Big)$$
$$\ll q^n \sum_{k\geq 1} \frac{1}{|P|^k} + O\Big(\frac{n}{\deg P}\Big) \ll O(q^n). \ \blacksquare$$

LEMMA 3.4.

$$\deg\Big(\prod_{n<\deg P\leq n+\deg f_d} P^{\alpha_P(n)}\Big) = O(q^n).$$

Proof. By Lemma 3.3 for sufficiently large n and P a prime such that $\deg P > n$, we have $\alpha_P(n) \leq q^n \frac{\rho_f(P)}{|P|-1} + O\big(\frac{n}{\deg P}\big) \ll \rho_f(P) + O(1) \ll 1$. Using the Prime Polynomial Theorem, we have

$$\deg\Big(\prod_{n<\deg P\leq n+\deg f_d} P^{\alpha_P(n)}\Big) = \sum_{n<\deg P\leq n+\deg f_d} \alpha_P(n)\deg P$$
$$\ll \sum_{n<\deg P\leq n+\deg f_d} \deg P = \sum_{k=n+1}^{n+\deg f_d} \sum_{\deg P=k} k$$
$$\ll \sum_{k=n+1}^{n+\deg f_d} k\frac{q^k}{k} = \frac{q^{n+\deg f_d+1} - q^{n+1}}{q-1} \ll q^n. \ \blacksquare$$

PROPOSITION 3.5. *Denote* $R_f(n) = \prod_{\deg P\leq n+\deg f_d} P^{\alpha_P(n)}$. *Then*

$$\deg R_f(n) = nq^n + O(q^n).$$

Proof. Let us assume first that f is separable. We will handle the inseparable case at the end of the proof. We note that we may ignore the $O(1)$ "bad" primes (as defined in Lemma 3.3) with an error term of $O(q^n)$ and by Lemma 3.4 we can ignore the primes with $n < \deg P \leq n + \deg f_d$ with the same error term. Thus by Lemma 3.3 we obtain

$$\deg R_f(n) = \sum_{\deg P\leq n} \alpha_P(n)\deg P + O(q^n) \tag{3.4}$$
$$= \sum_{\deg P\leq n} q^n \frac{\rho_f(P)}{|P|-1}\deg P + \sum_{\deg P\leq n} O(n) + O(q^n).$$

We bound the error term using the Prime Polynomial Theorem:

$$\sum_{\deg P\leq n} n = n\sum_{k=1}^{n}\sum_{\deg P=k} 1 \ll n\sum_{k=1}^{n}\frac{q^k}{k} \ll q^n. \tag{3.5}$$

Now to estimate $\sum_{\deg P\leq n} q^n \frac{\rho_f(P)}{|P|-1}\deg P$ we will use the Chebotarev Density Theorem in function fields [FJ08, Proposition 7.4.8]. First we introduce

some notation and recall some terminology. Let $E/\mathbb{F}$ be the splitting field of f and denote by G the Galois group of f. For each prime P of $\mathbb{F}_q[T]$ unramified in $E/\mathbb{F}_q(T)$ the *Frobenius class* of P is defined to be

$$\left(\frac{E/\mathbb{F}_q(T)}{P}\right) = \big\{\sigma \in G : \exists \mathfrak{P}/P \text{ prime of } E \text{ such that } x^{|P|} \equiv \sigma(x) \pmod{\mathfrak{P}} \text{ for all } x \text{ with } v_{\mathfrak{P}}(x) \geq 0\big\}.$$

It is a conjugacy class in G. Denote by S the set of all conjugacy classes in G. Given $C \in S$, we set

$$\pi_C(n) = \left|\left\{P \text{ prime} : \deg P = n,\ \left(\frac{E/\mathbb{F}_q(T)}{P}\right) = C\right\}\right|.$$

Let $K = \mathbb{F}_{q^v}$ be the algebraic closure of $\mathbb{F}_q$ in E. We have $G_0 := \mathrm{Gal}(K/\mathbb{F}_q) = \langle\phi\rangle$, where $\phi(x) = x^q$ is the q-Frobenius. Denote the restriction of automorphisms map by $\varphi : G \to G_0$. Since $G_0 = \langle\phi\rangle$ is cyclic and in particular abelian, for all $C \in S$ we have $\varphi(C) = \{\phi^{n_C}\}$ for some $n_C \in \mathbb{Z}$. Define

$$S_k := \{C \in S : \varphi(C) = \{\phi^k\}\}.$$

Now the Chebotarev Density Theorem in function fields says that

$$\pi_C(k) = \begin{cases} v\dfrac{|C|}{|G|}\,\dfrac{q^k}{k} + O\left(\dfrac{q^{k/2}}{k}\right), & C \in S_k, \\ 0, & C \notin S_k. \end{cases}$$

Since the Galois group acts on the roots of f, we can define $\mathrm{Fix}(C)$ to be the number of roots fixed by any element in the conjugacy class C (this number is the same for all $\sigma \in C$). Assuming P is unramified in $E/\mathbb{F}_q(T)$, we have $\rho_f(P) = \mathrm{Fix}\big(\big(\frac{E/\mathbb{F}_q(T)}{P}\big)\big)$. In the calculations throughout the rest of the proof, summation over P denotes summation over the primes $P \nmid \mathrm{Res}_X\big(f, f'\big)$ if f is separable and over $P \nmid \mathrm{Res}(f, \frac{\partial f}{\partial T})$ otherwise. If f is separable this conditions ensures that P is unramified in $E/\mathbb{F}_q(T)$. As we observed above, the excluded primes contribute $O(q^n)$.

We have

$$\begin{aligned} \sum_{\deg P = k} \frac{\rho_f(P)}{q^k - 1} k &= \sum_{\deg P = k} \frac{\mathrm{Fix}\big(\big(\frac{E/\mathbb{F}_q(T)}{P}\big)\big)}{q^k - 1} k = \sum_{C \in S} \frac{\mathrm{Fix}(C)\pi_C(k)}{q^k - 1} k \\ &= \sum_{C \in S_k} \left[\frac{\mathrm{Fix}(C) q^k}{(q^k - 1)k}\,\frac{v|C|}{|G|} k + O(q^{-k/2})\right] \\ &= \sum_{C \in S_k} v\frac{\mathrm{Fix}(C)|C|}{|G|} + O(q^{-k/2}) \end{aligned}$$

and therefore

$$\begin{aligned}(3.6)\qquad \sum_{\deg P\le n} q^n\frac{\rho_f(P)}{q^k-1}k &= q^n\sum_{k=1}^{n}\sum_{\deg P=k}\frac{\rho_f(P)}{q^k-1}k\\ &= q^n\sum_{k=1}^{n}\Big[\sum_{C\in S_k} v\frac{\mathrm{Fix}(C)|C|}{|G|}+O(q^{-k/2})\Big]\\ &= q^n\sum_{k=1}^{n}\sum_{C\in S_k} v\frac{\mathrm{Fix}(C)|C|}{|G|}+O(q^n)\\ &= q^n\sum_{l=1}^{\lfloor n/v\rfloor} v\sum_{C\in S}\frac{\mathrm{Fix}(C)|C|}{|G|}+O(q^n).\end{aligned}$$

Using Burnside's lemma [Rot95, Theorem 3.22] and the transitivity of G (which is a consequence of f being irreducible) we get

$$1=\sum_{\sigma\in G}\frac{\mathrm{Fix}(\sigma)}{|G|}=\sum_{C\in S}\frac{\mathrm{Fix}(C)|C|}{|G|}.$$

Plugging this into (3.6) and recalling (3.4), (3.5) we obtain

$$\begin{aligned}\deg R_f(n) &= \sum_{\deg P\le n} q^n\frac{\rho_f(P)}{|P|-1}\deg P+O(q^n)=q^n\sum_{l=1}^{\lfloor n/v\rfloor} v+O(q^n)\\ &= nq^n+O(q^n).\end{aligned}$$

This completes the proof in the separable case. To handle the case when f is inseparable we again do the same calculations (using Lemma 3.3(iii) this time):

$$\begin{aligned}\deg R_f(n) &= \sum_{\deg P\le n}\alpha_P(n)\deg P+O(q^n)\\ &= \sum_{\deg P\le n} q^n\frac{\rho_f(P)}{|P|}\deg P+\sum_{\deg P\le n}O(n)+O(q^n).\end{aligned}$$

We can replace $\rho_f(P)$ with $\rho_h(P)$, where h is the polynomial given by Lemma 2.10. Since h is separable, the argument above (with f replaced by h) yields

$$\deg R_f(n)=\sum_{\deg P\le n} q^n\frac{\rho_h(P)}{|P|}+O(q^n)=nq^n+O(q^n).\ \blacksquare$$

We are now ready to prove the upper bound on $\deg L_f(n)$.

Proof of Theorem 1.3. Using Lemma 3.2 we have $c_f\alpha_P(n)\le\beta_P(n)$ for all P prime. We deduce

$$\begin{aligned}
\deg L_f(n) &= c_f \deg P_f(n) - (c_f \deg P_f(n) - \deg L_f(n)) \\
&= c_f dnq^n - \sum_P (c_f \alpha_P(n) - \beta_P(n)) \deg P \\
&\le c_f dnq^n - \sum_{\deg P \le n + \deg f_d} (c_f \alpha_P(n) - \beta_P(n)) \deg P \qquad (3.7) \\
&= c_f dnq^n - c_f \sum_{\deg P \le n + \deg f_d} \alpha_P(n) \deg P \\
&\quad + \sum_{\deg P \le n + \deg f_d} \beta_P(n) \deg P \\
&= c_f (d-1) nq^n + \sum_{\deg P \le n + \deg f_d} \beta_P(n) \deg P + O(q^n), \qquad (3.8)
\end{aligned}$$

where (3.7) comes from removing negative terms and (3.8) from Proposition 3.5 and Lemma 3.1.

It remains to prove that $\sum_{\deg P \le n + \deg f_d} \beta_P(n) \deg P = O(q^n)$. From Lemma 3.3(i) and the Prime Polynomial Theorem we get

$$\begin{aligned}
&\sum_{\deg P \le n + \deg f_d} \beta_P(n) \deg P \qquad (3.9) \\
&\qquad \ll \sum_{\deg P \le n + \deg f_d} \frac{n}{\deg P} \deg P = \sum_{\deg P \le n + \deg f_d} n \ll q^n. \ \blacksquare
\end{aligned}$$

4. Heuristic justification for Conjecture 1.1. Throughout this section we maintain the assumption that $f = \sum_{i=0}^{d} f_i X^i \in \mathbb{F}_q[T][X]$ is irreducible. Next we want to study the difference $\deg L_f(n) - c_f(d-1)nq^n$ and we will do so by relating it to a slightly different quantity defined by (4.1) below. To this end let us define an equivalence relation on M_n by

$$Q_1 \sim Q_2 \iff f(X + Q_1 - Q_2) = f(X) \iff Q_1 - Q_2 \in V_f.$$

We note that for sufficiently large n the size of each equivalence class is $|V_f|$ and the number of such classes is $c_f q^n$. Consider

$$S_f(n) = \left| \left\{ P \in \mathbb{F}_q[T] \text{ prime} : \begin{array}{l} \deg P > n + \deg f_d, \\ \exists\, Q_1 \nsim Q_2 \text{ in } M_n \text{ with } P \mid f(Q_1), f(Q_2) \end{array} \right\} \right|. \qquad (4.1)$$

Note that the bottom condition in (4.1) is equivalent to $\beta_P(n) \ne c_f \alpha_P(n)$ (the negation of this condition is that P occurs precisely in all $f(Q)$ where Q runs over a single equivalence class of $\sim$, equivalently $|V_f| \beta_P(n) = \alpha_P(n)$),

hence

$$S_f(n) = \sum_{\substack{\deg P > n + \deg f_d \\ c_f \alpha_P(n) \neq \beta_P(n)}} 1. \tag{4.2}$$

PROPOSITION 4.1. *Let $f \in \mathbb{F}_q[T][X]$ be irreducible with $\deg_X f = d \geq 2$. Then we have the following:*

(i) $S_f(n) \ll q^n$.
(ii) *Conjecture* 1.1 *is equivalent to* $S_f(n) = o(q^n)$.

Proof. The first part follows immediately from the definition of $S_f(n)$, since there are q^n possible values of $f(Q)$ and each has $O(1)$ prime factors satisfying $\deg P > n + \deg f_d$. It remains to prove the second part.

In the main calculation in the proof of Theorem 1.3 there is only one inequality, namely (3.7):

$$c_f dnq^n - \sum_P (c_f \alpha_P(n) - \beta_P(n)) \deg P \leq c_f dnq^n - \sum_{\deg P \leq n + \deg f_d} (c_f \alpha_P(n) - \beta_P(n)) \deg P;$$

equivalently

$$- \sum_{\deg > n + \deg f_d} (c_f \alpha_P(n) - \beta_P(n)) \deg P \leq 0.$$

Since the RHS in Conjecture 1.1 is $\gg nq^n$, the conjecture is now seen to be equivalent to

$$\sum_{\deg > n + \deg f_d} (c_f \alpha_P(n) - \beta_P(n)) = o(nq^n).$$

Hence it suffices to prove that

$$S_f(n) = o(q^n) \iff \sum_{\deg P > n + \deg f_d} (c_f \alpha_P(n) - \beta_P(n)) \deg P = o(nq^n). \tag{4.3}$$

For $P \in \mathbb{F}_q[T]$ prime, set

$$\delta_P(n) = \begin{cases} 1 & \text{if } c_f \alpha_P(n) \neq \beta_P(n), \\ 0 & \text{otherwise.} \end{cases}$$

Note that for n sufficiently large, if $\deg P > dn + \deg f_d$ then $c_f \alpha_P(n) = \beta_P(n) = 0$. Hence, for n sufficiently large, using (4.2) we have

$$S_f(n) = \sum_{n + \deg f_d < \deg P \leq dn + \deg f_d} \delta_P(n).$$

From Lemma 3.3(ii–iii) and Proposition 2.9, for sufficiently large n and P not one of the $O(1)$ bad primes, if $\deg P > n$ then $\alpha_P(n) \ll 1$, so

$$\begin{aligned}(4.4)\qquad \sum_{n+\deg f_d<\deg P} &(c_f\alpha_P(n)-\beta_P(n))\deg P\\ &\ll \sum_{n+\deg f_d<\deg P\le dn+\deg f_d} \delta_P(n)\deg P\\ &\ll \sum_{n+\deg f_d<\deg P\le dn+\deg f_d} \delta_P(n)n = S_f(n)n.\end{aligned}$$

On the other hand, once again using the fact that $c_f\alpha_P(n) = \beta_P(n)$ if $\deg P > dn + \deg f_d$, we get

$$\begin{aligned}(4.5)\qquad \sum_{n+\deg f_d<\deg P} &(c_f\alpha_P(n)-\beta_P(n))\deg P\\ &\ge \sum_{n+\deg f_d<\deg P\le dn+\deg f_d} \delta_P(n)\deg P\\ &\ge \sum_{n+\deg f_d<\deg P\le dn+\deg f_d} \delta_P(n)n = S_f(n)n.\end{aligned}$$

From (4.4) and (4.5) the equivalence (4.3) follows. ∎

Heuristically one can estimate $S_f(n)$ by arguing that the "probability" that for $Q_1 \nsim Q_2$ in M_n the values $f(Q_1), f(Q_2)$ share a prime factor satisfying $\deg P > n + \deg f_d$ is (by the Prime Polynomial Theorem)

$$\sum_{m=n+\deg f_d+1}^{\infty} \sum_{\deg P=m} \frac{1}{|P|^2} = \sum_{m=n+\deg f_d+1}^{\infty} \frac{1}{mq^m} = O\Big(\frac{1}{nq^n}\Big).$$

Since there are $O(q^{2n})$ pairs Q_1, Q_2, we expect

$$S_f(n) \ll \frac{1}{nq^n}\cdot q^{2n} \ll \frac{q^n}{n} = o(q^n).$$

By Proposition 4.1 this is equivalent to Conjecture 1.1.

At this point one may ask whether the equivalence relation $\sim$ is really the dominant source of pairs $Q_1, Q_2 \in M_n$ with $f(Q_1) = f(Q_2)$, otherwise we would definitely not have $S_f(n) = o(q^n)$. Over the integers it cannot happen that $f(n_1) = f(n_2)$ for $n_1 \ne n_2 \in \mathbb{Z}$ sufficiently large (since a polynomial is a monotone function for sufficiently large arguments), but in finite characteristic this is less obvious. Nevertheless it is guaranteed by the following proposition, which will be needed for the proof of Theorem 1.8 as well.

Proposition 4.2.

$$|\{(Q_1,Q_2) : Q_1 \nsim Q_2 \text{ in } M_n,\ f(Q_1) = f(Q_2)\}| = O(q^{n/2}).$$

Before we prove Proposition 4.2, we need a couple of auxiliary results.

LEMMA 4.3. *Let $g(X,Y) \in \mathbb{F}_q[T][X,Y]$ be a fixed irreducible polynomial with $\deg_{X,Y} g \geq 2$. The number of pairs $Q_1, Q_2 \in M_n$ such that $g(Q_1,Q_2) = 0$ is $O(q^{n/2})$.*

Proof. If $\deg_X g = \deg_Y g = 1$ then $g = AXY + BX + CY + D$, $A \neq 0$, $B, C, D \in \mathbb{F}_q[T]$ and one cannot have $g(Q_1,Q_2) = 0$ for $Q_1, Q_2 \in M_n$, $n > \max(\deg B, \deg C, \deg D)$ since the term AQ_1Q_2 has higher degree than the other terms. Hence assume without loss of generality that $\deg_Y g \geq 2$, otherwise switch X and Y. It follows from the quantitative Hilbert Irreducibility Theorem in function fields [BSE21, Theorem 1.1] that for all but $O(q^{n/2})$ polynomials $Q_1 \in M_n$, the polynomial $g(Q_1, Y) \in \mathbb{F}_q[T][Y]$ is irreducible of degree ≥ 2 and therefore has no root $Q_2 \in M_n$. For the remaining $O(q^{n/2})$ values of Q_1 there are at most $\deg_Y g = O(1)$ roots for each Q_1, so overall there are $O(q^{n/2})$ pairs $Q_1, Q_2 \in M_n$ with $g(Q_1,Q_2) = 0$. ∎

LEMMA 4.4. *Assume that the polynomial $f(X) - f(Y) \in \mathbb{F}_q[T][X,Y]$ is divisible by $aX + bY + c$ for some $a, b, c \in \mathbb{F}_q[T]$ with $b \neq 0$. Then*

(i) *$\zeta := -a/b$ satisfies $\zeta^d = 1$ and $\zeta \in \mathbb{F}_q$;*
(ii) *if $\zeta = 1$ then $c/b \in V_f$.*

Proof. The divisibility assumption implies $f(X) = f\big(-\frac{a}{b}X - \frac{c}{b}\big)$. Comparing the coefficients at X^d gives $\zeta^d = 1$. Since $\zeta \in \mathbb{F}_q(T)$ and $\mathbb{F}_q$ is algebraically closed in $\mathbb{F}_q(T)$, we have $\zeta \in \mathbb{F}_q$, which establishes (i). If $\zeta = -1$ the above identity becomes $f(X) = f(X - c/b)$, i.e. $-c/b \in V_f$. Since V_f is a vector space, we obtain (ii). ∎

Proof of Proposition 4.2. Write $f(X) - f(Y) = c\prod_{i=1}^m g_i(X,Y)$ with $c \in \mathbb{F}_q[T]$ and each $g_i \in \mathbb{F}_q[T][X,Y]$ irreducible with $\deg_{X,Y} g_i \geq 1$. It is enough to show that for each factor $g_i(X,Y)$ there are $\ll q^{n/2}$ pairs $Q_1 \nsim Q_2$ in M_n with $g_i(Q_1,Q_2) = 0$. For a given i, if $\deg_{X,Y} g_i \geq 2$, this follows from Lemma 4.3. If $\deg_{X,Y} g_i = 1$, write $g_i = aX + bY + c$, assume without loss of generality that $b \neq 0$ and denote $\zeta = -a/b$. By Lemma 4.4 we have $\zeta^d = 1$.

We distinguish two cases. If $\zeta \neq 1$ then $a \neq -b$ and the leading coefficient of $g_i(Q_1,Q_2)$ is $a + b \neq 0$ for any $Q_1, Q_2 \in M_n$ with $n > \deg c$, and therefore there are no such pairs with $g_i(Q_1,Q_2) = 0$. If on the other hand $\zeta = 1$ (i.e. $a = -b$) then by Lemma 4.4 we have $c/b \in V_f$ and $g_i(Q_1,Q_2) = aQ_1 + bQ_2 + c = 0$ is equivalent to $Q_1 + c/b = Q_2$, so $Q_1 \sim Q_2$ and there are no pairs $Q_1, Q_2 \in M_n$ with $g_i(Q_1,Q_2) = 0$ and $Q_1 \nsim Q_2$. ∎

REMARK 4.5. It is readily seen from the proof that if f is special in the sense of Definition 1.6 then we may put 0 on the RHS of Proposition 4.2 for n sufficiently large, since in this case $\deg_{X,Y} g_i = 1$ for all i.

5. The lower bound

LEMMA 5.1. *Let $P \in \mathbb{F}_q[T]$ be a prime such that*

(a) $\deg P > \deg f_d + n$;
(b) $P \nmid \operatorname{Res}(f, f')$ *if f is separable;*
(c) $P \nmid \operatorname{Res}\left(f, \frac{\partial f}{\partial T}\right)$ *if f is inseparable.*

Denote

$$B_i = B_i(P) := |\{Q \in M_n : P^i \mid f(Q) \neq 0\}|.$$

Then

(i) $B_1 \leq d$;
(ii) $B_i \leq B_1$ *for* $i \geq 1$;
(iii) $B_i = 0$ *for* $i > d$.

Proof. Observe that since $n + \deg f_d < \deg P^i$, M_n is mapped injectivly into $\mathbb{F}_q[T]/P^i$ by $Q \mapsto f(Q) \bmod P^i$.

(i) By the observation above, $B_1 \leq \rho_f(P) \leq \deg(f \bmod P) = d$, so $B_1 \leq d$.

(ii) If f is separable then Lemma 2.4 combined with assumption (b) applied iteratively with $\alpha = 1, \ldots, i-1$ implies that each root of f modulo P has a unique lift to a root modulo P^i. Combined with the observation above, this implies $B_i \leq B_1$. If f is inseparable then by Lemma 2.8 and assumption (c) we have $B_i \leq \rho_f(P^i) = 0 \leq B_1$ for $i > 1$.

(iii) Finally, suppose that $i > d$. If there exists $Q \in \mathbb{F}_q[T]$ such that $P^i \mid f(Q) \neq 0$, then $\deg f(Q) \leq dn + \deg f_d$, while $dn + \deg f_d < i \deg P = \deg P^i$, a contradiction. Hence $B_i = 0$ in this case. ∎

Proof of Theorem 1.4(i). From Lemma 5.1 we deduce that for all but $O(1)$ primes $P \in \mathbb{F}_q[T]$ (note that by Lemma 2.7 conditions (b–c) of Lemma 5.1 are satisfied for all but $O(1)$ primes) such that $\deg P > n + \deg f_d$ we have

$$\alpha_P(n) = \sum_{i \geq 1} \sum_{\substack{Q \in M_n \\ P^i \mid f(Q)}} 1 = \sum_{\substack{i \geq 1 \\ B_i > 0}} B_i \leq d \sum_{\substack{i \geq 1 \\ B_i > 0}} 1 \leq d\beta_P(n).$$

Now using Lemma 3.1 and Proposition 3.5 we obtain

$$\begin{aligned}(d-1)nq^n \lesssim \deg \frac{P_f(n)}{R_f(n)} &= \sum_{\deg P > n + \deg f_d} \alpha_P(n) \deg P \\ &\leq d \sum_{\deg P > n + \deg f_d} \beta_P(n) \deg P \leq d \deg L_f(n),\end{aligned}$$

hence $\deg L_f(n) \gtrsim \frac{d-1}{d} nq^n$ as required. ∎

REMARK 5.2. There exist polynomials f for which the lower bound given by Theorem 1.4(i) and the upper bound given by Theorem 1.3 are equal.

Indeed, if $|V_f| = d$ (as in Example 2.2), then combining Theorems 1.3 and 1.4(ii) gives $\deg L_f(n) \sim \frac{d-1}{d} nq^n$ as $n \to \infty$. Hence the lower bound is tight in this case.

To prove the second part of Theorem 1.4 we first need the following

LEMMA 5.3. *Assume $p \nmid d$ or $f_d \nmid f_{d-1}$. Let n be sufficiently large and $P \in \mathbb{F}_q[T]$ a prime such that $\deg P > \deg f_d + n$. Then in the notation of Lemma 5.1 we have $B_1(P) \le d-1$.*

Proof. Assume by way of contradiction that there exist distinct $Q_1, \dots, Q_d$ such that $P \mid f(Q_1), \dots, f(Q_d)$. The assumption $\deg P > \deg f_d + n$ implies that $Q_i \bmod P$ are also distinct and therefore

$$f(X) \equiv f_d \prod_{j=1}^{d} (x - Q_j) \pmod{P}.$$

Comparing the coefficients of X^{d-1} we obtain

$$f_{d-1} \equiv -f_d \sum_{j=1}^{d} Q_j \pmod{P}, \quad \text{i.e.} \quad f_{d-1} + f_d \sum_{j=1}^{d} Q_j \equiv 0 \pmod{P}.$$

Thus $P \mid f_{d-1} + f_d \sum_{j=1}^{d} Q_j$ and if $f_{d-1} + f_d \sum_{j=1}^{d} Q_j \neq 0$ then we would have $\deg P \le n + \deg f_d$ (if $n \ge f_{d-1}$), a contradiction.

Now we observe that as $Q_1, \dots, Q_d$ are monic, if $\deg \sum_{j=1}^{d} Q_j < n$ then $p \mid d$. Therefore for a sufficiently large n if $f_{d-1} + f_d \sum_{j=1}^{d} Q_j = 0$ then $p \mid d$ and additionally $f_d \sum_{j=1}^{d} Q_j = -f_{d-1}$, hence $f_d \mid f_{d-1}$. This contradicts our initial assumption, and thus we have reached a contradiction in all cases. ∎

Proof of Theorem 1.4(ii). From Lemma 5.3 we deduce that for $P \in \mathbb{F}_q[T]$ prime such that $\deg P > n + \deg f_d$ we have $B_1(P) \le d-1$. The rest of the proof is now the same as for Theorem 1.4(i), with the inequality $\alpha_P(n) \le d\beta_P(n)$ replaced by $\alpha_P(n) \le (d-1)\beta_P(n)$, which has the effect of replacing the constant $\frac{d-1}{d}$ by 1 in the conclusion. ∎

6. The radical of $L_f(n)$. We assume for simplicity of exposition that f is irreducible, although this assumption can be relaxed to f being squarefree in Theorem 1.5 and its proof, which we give in the present section. Recall that we denote $\ell_f(n) = \operatorname{rad} L_f(n)$ and observe that

$$\deg \ell_f(n) = \sum_{\substack{P \\ \beta_P(n) > 0}} \deg P.$$

By Theorems 1.3 and 1.4(i) we have $nq^n \ll \deg L_f(n) \ll nq^n$, and the asymptotic $\deg \ell_f(n) \sim \deg L_f(n)$ (Theorem 1.5) is equivalent to the following.

Proposition 6.1.

$$\deg L_f(n) - \deg \ell_f(n) = o(nq^n).$$

Proof. Let us write

$$\ell_f(n) = \sum_{\substack{\deg P > n \\ \beta_P(n) > 0}} \deg P + \sum_{\substack{\deg P \le n \\ \beta_P(n) > 0}} \deg P,$$

$$L_f(n) = \sum_{\deg P > n} \beta_P(n) \deg P + \sum_{\deg P \le n} \beta_P(n) \deg P.$$

Now using (3.9) we obtain

$$\sum_{\substack{\deg P \le n \\ \beta_P(n) > 0}} \deg P \le \sum_{P \le n + \deg f_d} \beta_P(n) \deg P = O(q^n)$$

and therefore

$$\ell_f(n) = \sum_{\substack{\deg P > n \\ \beta_P(n) > 0}} \deg P + O(q^n), \qquad L_f(n) = \sum_{\substack{\deg P > n \\ \beta_P(n) > 0}} \beta_P(n) \deg P + O(q^n).$$

Denote

$$S := \{Q \in M_n : \exists P \text{ prime with } \deg P > n,\ P^2 \mid f(Q)\}.$$

We have

$$0 \le \deg L_f(n) - \deg \ell_f(n) = \sum_{\deg P > n} \beta_P(n) \deg P - \sum_{\substack{\deg P > n \\ \beta_P(n) > 0}} \deg P + O(q^n)$$

$$\le \sum_{\substack{\deg P > n \\ \beta_P(n) \ge 2}} \beta_P(n) \deg P + O(q^n) \le \sum_{Q \in S} \deg f(Q) + O(q^n) \ll |S| n + q^n.$$

It remains to prove that $|S| = o(q^n)$; but by [Poo02, Lemma 7.1], the set

$$W = \{\deg Q \le n : \exists P \text{ prime with } \deg P > n/2,\ P^2 \mid f(Q)\}$$

is of size $o(q^n)$ (this is the key ingredient of the proof; see also [Lan15, Proposition 3.3] and [Car21, (4.7)] for refinements of this statement). Since $S \subset W$, we have $|S| = o(q^n)$ and we are done. ∎

As noted above, this concludes the proof of Theorem 1.5.

Remark 6.2. Assuming the ABC conjecture, one can prove that $\log \ell_f(N) \sim \log L_f(N)$ over $\mathbb{Z}$. The argument is essentially the same as in the proof of Theorem 1.5 above, using the fact that under ABC one has

$$|\{n \le N : \exists p \text{ prime with } p > N,\ p^2 | f(n)\}| = o(N),$$

which follows from [Gra98, Theorem 8].

7. Asymptotics of $L_f(n)$ for special polynomials. In the present section we assume that $f = \sum_{i=0}^{d} f_i X^i \in \mathbb{F}_q[T][X]$, $\deg f = d \geq 2$ is irreducible and special in the sense of Definition 1.6. We will show that $\deg L_f(n) \sim c_f n q^n$ as $n \to \infty$, thus proving Theorem 1.8. As in Section 4, we consider the equivalence relation on M_n given by $Q_1 \sim Q_2 \Leftrightarrow Q_1 - Q_2 \in V_f$ and the quantity $S_f(n)$ defined by (4.1).

By the definition of a special polynomial we may write

$$f(X) - f(Y) = \prod_{i=1}^{d} (a_i X + b_i Y + c_i), \qquad a_i, b_i, c_i \in \mathbb{F}_q[T]. \tag{7.1}$$

Comparing the degrees in X and Y shows that $a_i, b_i \neq 0$. Comparing the coefficients of X^d and Y^d in (7.1) we see that $\deg a_i, \deg b_i \leq \deg f_d$. Therefore if $Q_1, Q_2 \in M_n$ with $n > \deg c_i$, we have

$$\deg(a_i Q_1 + b_i Q_2 + c_i) \leq n + \deg f_d. \tag{7.2}$$

Let $Q_1, Q_2 \in M_n$ with $Q_1 \nsim Q_2$ and P a prime with $\deg P > n + \deg f_d$ such that $P \mid f(Q_1), f(Q_2)$. Then in particular $P \mid f(Q_1) - f(Q_2)$ and by (7.1) we have $P \mid a_i Q_1 + b_i Q_2 + c_i \neq 0$ for some $1 \leq i \leq d$. By (7.2) and the condition $\deg P > n + \deg f_d$ we must have $a_i Q_1 + b_i Q_2 + c_i = 0$ and therefore by (7.1) we have $f(Q_1) = f(Q_2)$. By Proposition 4.2 combined with Remark 4.5 this is impossible for n sufficiently large. Hence $S_f(n) = 0$ for n sufficiently large and by Proposition 4.1 it follows that Conjecture 1.1 holds for f, concluding the proof of Theorem 1.8.

8. Classification of special polynomials. In the present section we classify the special polynomials (Definition 1.6) over an arbitrary unique factorization domain (UFD) R, establishing Theorem 1.10. We denote by p the characteristic of R (as R is a domain, p is zero or a prime) and by K its field of fractions. Also, if $p > 0$, we will denote by $\mathbb{F}_p$ its prime subfield. Note that as R is a UFD, so are the polynomial rings $R[X], R[X,Y]$. For a polynomial $g \in R[X,Y]$ we denote by $\deg g$ its total degree. Special polynomials $f \in R[X]$ were defined in Definition 1.6.

LEMMA 8.1. *Let $f \in R[X]$ be a special polynomial of degree d. Then we may write*

$$f(X) - f(Y) = \prod_{i=1}^{d} (a_i X + b_i Y + c_i), \qquad a_i, b_i \neq 0,\ c_i \in R. \tag{8.1}$$

Proof. This is immediate from the definition, except for the condition $a_i, b_i \neq 0$ which follows by comparing the degrees in X and Y in (8.1). ∎

LEMMA 8.2. *Let $f = \sum_{i=0}^{d} f_i X^i \in R[X]$ be a special polynomial of degree d.*

(i) *If $p = 0$ then*

$$f(X) - f(Y) = f_d \prod_{j=1}^{d} (X - \zeta^j Y - b^{(j)}),$$

where $\zeta \in R$ is a primitive dth root of unity and $b^{(j)} \in K$.

(ii) *If $p > 0$, write $d = p^l m$ with $(m, p) = 1$. Then*

$$f(X) - f(Y) = f_d \prod_{i=1}^{p^l} \prod_{j=1}^{m} (X - \zeta^j Y - b_i^{(j)}),$$

where $\zeta \in R$ is a primitive mth root of unity and $b_i^{(j)} \in K$.

Proof. For brevity we only treat the case $p > 0$, the case $p = 0$ being similar to the case $p > 0$, $l = 0$. Write $f(X) - f(Y) = \prod_{i=1}^{d} (a_i X + b_i Y + c_i)$ as in Lemma 8.1. Note that

$$\prod_{i=1}^{d} (a_i X + b_i Y + c_i) = \prod_{i=1}^{d} (a_i X + b_i Y) + A[X, Y],$$

where $A[X, Y] \in R[X, Y]$, $\deg A < d$. Since $\prod_{i=1}^{d} (a_i X + b_i Y)$ is homogeneous of degree d, we have

$$f_d (X^d - Y^d) = \prod_{i=1}^{d} (a_i X + b_i Y) = \prod_{i=1}^{d} a_i \prod_{i=1}^{d} (X - \mu_i Y),$$

where $\mu_i = -b_i / a_i \in K$. Plugging in $Y = 1$ and noting that $\prod_{i=1}^{d} a_i = f_d$ (by comparing the coefficients of X^d) we obtain

$$X^d - 1 = \prod_{i=1}^{d} (X - \mu_i).$$

In particular, we see that all the dth roots of unity in the algebraic closure of K lie in K, and since R is integrally closed (because it is a UFD), they lie in R. Since $p = \mathrm{char}(K) > 0$, these are actually the mth roots of unity, where $d = mp^l$, $(m, p) = 1$. ∎

We are ready to prove the forward direction of Theorem 1.10.

Proposition 8.3. *Let $f \in R[X]$ be special. Then it has the form stated in Theorem* 1.10.

Proof. For brevity we only treat the case $p > 0$, the case $p = 0$ being similar to the case $p > 0$, $l = 0$ that we treat below (using part (i) of Lemma 8.2 instead of part (ii), and the fact that $V_f = \{0\}$; see Remark 1.2). Write $d = \deg f = p^l m$, $(m, p) = 1$. Denote by ζ a primitive mth root of

unity. From Lemma 8.2(ii) we have $\zeta \in R$ and

$$f(X) - f(Y) = f_d \prod_{i=1}^{p^l} \prod_{j=1}^{m} (X - \zeta^j Y - b_i^{(j)}), \quad b_i^{(j)} \in K. \tag{8.2}$$

Next consider the shifted polynomial $g(X) = f(X + w) \in K[X]$ with

$$w = \begin{cases} \dfrac{b_1^{(m-1)}}{1 - \zeta^{-1}}, & m > 1, \\ 0, & m = 1. \end{cases}$$

Note that g is also special but over the ring K instead of R (this is immediate from the definition) and moreover by the choice of w we have

$$g(\zeta X) = g(X),$$

because

$$\begin{aligned} g(X) - g(\zeta X) &= f(X + w) - f(\zeta X + w) \\ &= f_d \prod_{i=1}^{p^l} \prod_{j=1}^{m} \big(X + w - \zeta^j(\zeta X + w) - b_i^{(j)}\big) \end{aligned}$$

and if $m > 1$ then $X + w - \zeta^{m-1}(\zeta X + w) - b_1^{(m-1)} = 0$ (since $\zeta^m = 1$). If we can show that g has the form asserted by Theorem 1.10 except that the coefficients are in K instead of R, then $f(X) = g(X - w)$ would have the form asserted by the theorem. Replacing f with g, we see that it is enough to prove the assertion under the additional assumptions that $R = K$ is a field and $f(X) = f(\zeta X)$.

Thus from now on we assume that $R = K$ is a field and $f(X) = f(\zeta X) = f(\zeta^2 X) = \cdots = f(\zeta^{m-1} X)$. Next let us prove that for all j the multisets

$$S_j := \{b_i^{(j)} : 1 \le i \le p^l\}$$

are the same for all $1 \le j \le m$. To this end let us identify the indices $1 \le j \le m$ with their corresponding residues in $\mathbb{Z}/m$ and observe that by (8.2) we have

$$\begin{aligned} \prod_{j \bmod m} \prod_{b \in S_j} (X - \zeta^j Y - b) &= f(X) - f(Y) = f(X) - f(\zeta^k Y) \\ &= \prod_{j \bmod m} \prod_{b \in S_j} (X - \zeta^{j+k} Y - b) \\ &= \prod_{j \bmod m} \prod_{b \in S_{j-k}} (X - \zeta^j Y - b) \end{aligned} \tag{8.3}$$

for each $k \in \mathbb{Z}/m$. From the first equality in (8.3) we see that the number of times an element b appears in the multiset S_j equals the multiplicity of the

factor $X - \zeta^j Y - b$ in the factorization of $f(X) - f(Y)$ (recall that $R[X, Y]$ is a UFD), and from the last equality it also equals the number of times b appears in S_{j-k}. Hence the multisets $S_1, \dots, S_m$ are all equal and we may rewrite (8.2) as

$$f(X) - f(Y) = f_d \prod_{i=1}^{p^l} \prod_{j \bmod m} (X - \zeta^j Y - b_i) = f_d \prod_{b \in S_1} \prod_{j \bmod m} (X - \zeta^j Y - b), \tag{8.4}$$

where $b_i = b_i^{(1)}$.

Let us prove the following properties of the multiset $S_1 = \{b_1, \dots, b_m\}$:

(1) The underlying set of S_1 is $V_f := \{b \in R : f(X + b) = f(X)\}$. It is an $\mathbb{F}_p$-linear subspace of R, with $|V_f| = p^v$ for some $0 \le v \le l$.
(2) Multiplication by ζ permutes S_1 as a multiset.
(3) V_f is furthermore an $\mathbb{F}_p(\zeta)$-linear subspace of R.
(4) All elements of S_1 have the same multiplicity p^{l-v}.

For (1) observe that by (8.4) we have

$$b \in S_1 \iff X - Y - b \,|\, f(X) - f(Y) \iff f(X + b) = f(X),$$

hence V_f is the underlying set of S_1. Arguing as in the proof of Lemma 2.1 one sees that V_f is an $\mathbb{F}_p$-linear subspace of R. Since $|V_f| \le |S_1| = p^l$, we see that $|V_f| = p^v$ with $0 \le v \le l$. Replacing Y with ζY in (8.4) and using $f(Y) = f(\zeta Y)$ gives (2). Properties (1) and (2) imply (3).

To prove (4) observe that by (8.4) the multiplicity of $b \in V_f$ in S_1 equals the exponent of the factor $X - Y - b$ in the factorization of $f(X) - f(Y)$, but since $f(Y + b) = f(Y)$, it also equals the exponent of $X - Y$ in the factorization of $f(X) - f(Y)$ and is therefore independent of b. We have established properties (1)–(4).

Now assuming without loss of generality that $V_f = \{b_1, \dots, b_{p^v}\}$ we can rewrite (8.4) as

$$f(X) - f(Y) = f_d \prod_{i=1}^{p^v} \prod_{j=1}^{m} (X - \zeta^j Y - b_i)^{p^{l-v}} = f_d \prod_{i=1}^{p^v} ((X - b_i)^{mp^{l-v}} - b_i^{mp^{l-v}} Y^{mp^{l-v}}).$$

Setting $Y = 0$ we see that f has the form asserted in Theorem 1.10 with $A = 0$, $C = f(0)$. ∎

Finally, we prove the converse direction in Proposition 8.3.

PROPOSITION 8.4.

(i) *Assume that $p = 0$, K contains a primitive dth root of unity ζ and*

$$f = f_d(X + A)^d + C \in R[X]$$

with $A, C \in K$. Then f is special.

(ii) *Assume that $p > 0$, $d = p^l m$, $(m, p) = 1$ and K contains a primitive mth root of unity. Suppose*

$$f(X) = f_d \prod_{i=1}^{p^v} (X - b_i + A)^{mp^{l-v}} + C \in R[X],$$

where $A, C \in K$ and $V = \{b_1, \dots, b_{p^v}\}$, $0 \le v \le l$, (b_i distinct) is an $\mathbb{F}_p(\zeta)$-linear subspace of K. Then f is special.

Proof. (i) Denote by ζ a primitive dth root of unity. Then we have the factorization

$$f(X) - f(Y) = f_d \prod_{i=0}^{m-1} \big(X + A - \zeta^i(Y + A)\big)$$

and f is special.

(ii) Since R is a UFD, by Gauss's lemma $f(X) - f(Y)$ factors into linear polynomials over R if and only if it does over K. Hence we assume without loss of generality that $R = K$ is a field and $A, C, b_i \in R$. It is then immediate from the definition that $f(X)$ is special if and only if $\frac{1}{f_d}(f(X - A) - C)$ is, hence we assume without loss of generality that $A = C = 0$, $f_d = 1$, i.e. $f = g^{mp^{l-v}}$, where

$$g = \prod_{i=1}^{p^v} (X - b_i).$$

We now show that the following factorization holds:

$$f(X) - f(Y) = \prod_{i=1}^{p^v} \prod_{j=1}^{m} (X - \zeta^j Y + b_i)^{p^{l-v}}. \tag{8.5}$$

Since the Frobenius map is a ring homomorphism, this is equivalent to

$$g(X)^m - g(Y)^m = \prod_{i=1}^{p^v} \prod_{j=1}^{m} (X - \zeta^j Y + b_i). \tag{8.6}$$

Note that since V is an $\mathbb{F}_p(\zeta)$-linear subspace, the map $x \mapsto \zeta^{-j}(x + b_i)$ permutes V and we have

$$g(\zeta^{-j}(X + b_i)) = \zeta^{-jp^v} g(X)$$

and so $g(\zeta^{-j}(X+b_i))^m = g(X)^m$. It follows that $X - \zeta^j Y + b_i \mid g(X)^m - g(Y)^m$ and therefore the RHS of (8.6) divides the LHS. Since both sides of (8.6)

have the same degree mp^v and are monic in X, we must have an equality. This establishes (8.6) and therefore (8.5). ∎

Remark 8.5. Proposition 8.3 and its proof work under the weaker assumption that R is an integrally closed domain (not necessarily a UFD). Proposition 8.4 does require the UFD assumption, but it can be replaced with R being only integrally closed if one assumes that $f_d = 1$, because Gauss's lemma works for monic polynomials over any integrally closed domain.

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